Chapter 3 Multidimensional Random Variables

Section 4 Functions of Two Random Variables

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Outline

1. Functions of Two Random Variables
   - Discrete Random Variables
   - Discrete Examples
   - Continuous Random Variables
   - The Jacobian Transformation Method
   - Maximum and Minimum

Discrete Case: PMF of a Function

Let (X, Y) be a discrete random vector with joint PMF $p_{ij} = P(X = x_i, Y = y_j)$.
Let $Z = g(X, Y)$. Then Z is also a discrete random variable. Its possible values are $z_k = g(x_i, y_j)$.
The PMF of Z is obtained by summing the probabilities of all pairs (x_i, y_j) that map to z_k:

$$P(Z = z_k) = \sum_{g(x_i, y_j) = z_k} P(X = x_i, Y = y_j).$$

Convolution Formula for Discrete Variables

Theorem: Discrete Convolution

If X and Y are independent discrete random variables taking non-negative integers, with PMFs $P(X = i) = p_i$ and $P(Y = j) = q_j$, then for $Z = X + Y$:

$$P(Z = k) = \sum_{i=0}^{k} P(X = i, Y = k - i) = \sum_{i=0}^{k} p_i q_{k-i}, \quad k = 0, 1, 2, \ldots$$

This is the **discrete convolution formula**.

Example 1: Sum of Two Coin Tosses (Question)

Example

Let X and Y be independent, with $P(X = 0) = P(X = 1) = \frac{1}{2}$ and $P(Y = 0) = P(Y = 1) = \frac{1}{2}$.
Find the PMF of $Z = X + Y$.

Example 1: Sum of Two Coin Tosses (Solution)

Analysis: (X, Y) has 4 possible outcomes. The value of Z ranges from 0 to 2. We sum the probabilities from the joint PMF.

Solution

(X, Y)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$Z = X + Y$	0	1	1	2
Probability	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Therefore:

$$P(Z=0) = \frac{1}{4}, \quad P(Z=1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \quad P(Z=2) = \frac{1}{4}.$$

Example 2: Binomial Additivity (Question)

Example

Let $X \sim B(n_1, p)$ and $Y \sim B(n_2, p)$ be independent. Prove that $X + Y \sim B(n_1 + n_2, p)$.

Example 2: Binomial Additivity (Solution)

Analysis: Use the discrete convolution formula and the Vandermonde identity.

Proof

$$
\begin{aligned}
P(X+Y=k) &= \sum_{i=0}^{k} \binom{n_1}{i} p^i (1-p)^{n_1-i} \binom{n_2}{k-i} p^{k-i} (1-p)^{n_2-k+i} \\
&= p^k (1-p)^{n_1+n_2-k} \sum_{i=0}^{k} \binom{n_1}{i} \binom{n_2}{k-i} \\
&= \binom{n_1+n_2}{k} p^k (1-p)^{n_1+n_2-k}.
\end{aligned}
$$

Thus, $X+Y \sim B(n_1+n_2, p)$.

Example 3: Poisson Additivity (Question)

Example

Let $X \sim P(\lambda_1)$ and $Y \sim P(\lambda_2)$ be independent. Prove that $X + Y \sim P(\lambda_1 + \lambda_2)$.

Example 3: Poisson Additivity (Solution)

Analysis: Apply the convolution formula and use the binomial expansion.

Proof

$$
\begin{aligned}
P(X+Y=k) &= \sum_{i=0}^{k} \frac{\lambda_1^i}{i!} e^{-\lambda_1} \frac{\lambda_2^{k-i}}{(k-i)!} e^{-\lambda_2} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{i=0}^{k} \frac{k!}{i!(k-i)!} \lambda_1^i \lambda_2^{k-i} \\
&= \frac{(\lambda_1+\lambda_2)^k}{k!} e^{-(\lambda_1+\lambda_2)}.
\end{aligned}
$$

Example 4: Geometric Convolution (Question)

Example

Let $X \sim G(p)$ and $Y \sim G(p)$ be independent. Find the PMF of $Z = X + Y$.

Example 4: Geometric Convolution (Solution)

Analysis: For the geometric distribution, $X, Y \in \{1, 2, \dots\}$. Use the convolution formula for $k \geq 2$.

Solution

$$\begin{aligned} P(Z = k) &= \sum_{i=1}^{k-1} (1-p)^{i-1} p \cdot (1-p)^{k-i-1} p \\ &= p^2 \sum_{i=1}^{k-1} (1-p)^{k-2} = (k-1) p^2 (1-p)^{k-2}. \end{aligned}$$

This is the negative binomial distribution (waiting time for the 2nd success).

Remark: Binomial as a Sum of Bernoullis

Connection between $B(n, p)$ and $B(1, p)$

If $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d.) as $B(1, p)$ (Bernoulli), then by the additivity of Binomial distributions:

$$X_1 + X_2 + \cdots + X_n \sim B(n, p).$$

This shows that the Binomial distribution is just the sum of n independent Bernoulli trials.

General Method: CDF Approach

Steps to Find the Distribution of $Z = g(X, Y)$

Let (X, Y) be continuous with joint density $f(x, y)$.

1. Write the CDF of Z: $F_Z(z) = P(Z \leq z) = P(g(X, Y) \leq z)$.
2. Find the region $D_z = \{(x, y) : g(x, y) \leq z\}$.
3. Compute the double integral: $F_Z(z) = \iint_{D_z} f(x, y)\,dx\,dy$.
4. Differentiate to get the density: $f_Z(z) = F'_Z(z)$.

Density of a Sum: Convolution Formula

Let $Z = X + Y$. The CDF is:

$$F_Z(z) = \iint_{x+y \le z} f(x,y)\,dxdy = \int_{-\infty}^{+\infty} \int_{-\infty}^{z-x} f(x,y)\,dydx.$$

Differentiating with respect to z yields:

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, z-x)\,dx = \int_{-\infty}^{+\infty} f(z-y, y)\,dy.$$

If X and Y are independent, $f(x,y) = f_X(x) f_Y(y)$, then:

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx.$$

Example 5: Sum of Independent Normals (Question)

Example

Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ be independent. Find the distribution of $Z = X + Y$.

Example 5: Sum of Independent Normals (Solution)

Analysis: Use the continuous convolution formula. Combine the exponents and complete the square with respect to x.

Solution

The exponent is $-\frac{1}{2}\left[\frac{(x-\mu_1)^2}{\sigma_1^2}+\frac{(z-x-\mu_2)^2}{\sigma_2^2}\right]$.

Let $A=\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}=\frac{\sigma_1^2+\sigma_2^2}{\sigma_1^2\sigma_2^2}$. After algebra, it becomes:

$$-\frac{1}{2}A\left(x-\frac{\mu_1/\sigma_1^2+(z-\mu_2)/\sigma_2^2}{A}\right)^2-\frac{(z-(\mu_1+\mu_2))^2}{2(\sigma_1^2+\sigma_2^2)}.$$

Integrating over x yields $1/\sqrt{A}$, leading to:

$$f_Z(z)=\frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2+\sigma_2^2}}e^{-\frac{(z-(\mu_1+\mu_2))^2}{2(\sigma_1^2+\sigma_2^2)}}.$$

Thus, $Z\sim N(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2)$.

Example 6: Sum of Independent Exponentials (Question)

Example

Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$ be independent. Find the density of $Z = X + Y$.

Example 6: Sum of Independent Exponentials (Solution 1)

Analysis: For $z > 0$, $f_Z(z) = \int_0^z \lambda e^{-\lambda x} \mu e^{-\mu(z-x)} dx = \lambda\mu e^{-\mu z} \int_0^z e^{(\mu-\lambda)x} dx$.

Solution: Case 1

When $\lambda \neq \mu$:

$$\int_0^z e^{(\mu-\lambda)x} dx = \frac{e^{(\mu-\lambda)z} - 1}{\mu - \lambda}.$$

Thus,

$$f_Z(z) = \frac{\lambda\mu}{\mu - \lambda}(e^{-\lambda z} - e^{-\mu z}), \quad z > 0.$$

Example 6: Sum of Independent Exponentials (Solution 2)

Solution: Case 2

When $\lambda = \mu$:

$$\int_0^z e^0 dx = z.$$

Thus,

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0 \text{ (Erlang with } r = 2).$$

Solution: Case 3

When $\mu = 2\lambda$: Substitute into Case 1:

$$f_Z(z) = \frac{\lambda \cdot 2\lambda}{\lambda}(e^{-\lambda z} - e^{-2\lambda z}) = 2\lambda(e^{-\lambda z} - e^{-2\lambda z}), \quad z > 0.$$

Remark: Erlang and Gamma Distributions

Erlang Distribution

If $X_1, \ldots, X_r$ are i.i.d. Exp(λ), then $Z = X_1 + \cdots + X_r$ follows the Erlang distribution:

$$f_Z(z) = \frac{\lambda^r z^{r-1} e^{-\lambda z}}{(r-1)!}, \quad z > 0.$$

Gamma Distribution: Extending the factorial to the Gamma function $\Gamma(r)$ gives:

$$f_Z(z) = \frac{\lambda^r z^{r-1} e^{-\lambda z}}{\Gamma(r)}, \quad z > 0, r > 0.$$

Erlang is a special case of Gamma where r is an integer. Exponential is the case where $r = 1$.

Density of a Product: $Z = XY$

Transformation Method

Let $U = X$, $V = XY$. Then $x = u, y = v/u$. The Jacobian determinant is:

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1 & 0 \\ -v/u^2 & 1/u \end{vmatrix} = \frac{1}{u}.$$

The joint density is $f_{U,V}(u,v) = f(u, v/u)\frac{1}{|u|}$. Integrating out u gives the density of $Z = V$:

$$f_Z(z) = \int_{-\infty}^{+\infty} f\left(u, \frac{z}{u}\right) \frac{1}{|u|} du.$$

If X, Y are independent: $f_Z(z) = \int_{-\infty}^{+\infty} f_X(u) f_Y(z/u) \frac{1}{|u|} du$.

Example 7: Product of Two Uniforms (Question)

Example

Let X, Y be independent $U(0,1)$. Find the density of $Z = XY$.

Example 7: Product of Two Uniforms (Solution)

Analysis: For $0 < z < 1$, the constraints are $0 < u < 1$ and $0 < z/u < 1$, which means $z < u < 1$.

Solution

$$f_Z(z) = \int_z^1 1 \cdot 1 \cdot \frac{1}{u} du = \ln u \Big|_z^1 = -\ln z, \quad 0 < z \leq 1.$$

Outside $[0, 1]$, $f_Z(z) = 0$.

Density of a Quotient: $Z = X/Y$

For $Z = X/Y$, the CDF approach yields:

$$F_Z(z) = \int_0^{+\infty} \int_{-\infty}^{zy} f(x,y)\,dxdy + \int_{-\infty}^{0} \int_{zy}^{+\infty} f(x,y)\,dxdy.$$

Differentiating with respect to z:

$$f_Z(z) = \int_0^{+\infty} yf(zy,y)\,dy - \int_{-\infty}^{0} yf(zy,y)\,dy = \int_{-\infty}^{+\infty} |y| f(zy,y)\,dy.$$

If X, Y are independent:

$$f_Z(z) = \int_{-\infty}^{+\infty} |y| f_X(zy) f_Y(y)\,dy.$$

Example 8: Quotient of Two Exponentials (Question)

Example

Let $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$ be independent. Find the density of $Z = X/Y$.

Example 8: Quotient of Two Exponentials (Solution)

Analysis: For $z > 0$, use the quotient formula:

$$f_Z(z) = \int_0^{+\infty} y \cdot \lambda_1 e^{-\lambda_1 z y} \cdot \lambda_2 e^{-\lambda_2 y} dy.$$

Combine the exponents and use $\int_0^\infty y e^{-ay} dy = \frac{1}{a^2}$.

Solution

$$f_Z(z) = \lambda_1 \lambda_2 \int_0^\infty y e^{-(\lambda_1 z + \lambda_2) y} dy = \frac{\lambda_1 \lambda_2}{(\lambda_1 z + \lambda_2)^2}, \quad z > 0.$$

Outside this range, $f_Z(z) = 0$.

General Transformation: The Jacobian Method (Part 1)

The CDF method is intuitive but often requires complex integrals. For general transformations, we use the **Jacobian method**.

Methodology: Transformation

Let (X, Y) have joint density $f_{X,Y}(x, y)$. Consider a transformation to new variables (U, V):

$$U = g_1(X, Y), \quad V = g_2(X, Y).$$

Assume the inverse mapping exists and has continuous partial derivatives:

$$x = h_1(u, v), \quad y = h_2(u, v).$$

Goal: We want to find the joint density $f_{U,V}(u, v)$ of the new variables, and then the marginal density $f_U(u)$.

General Transformation: The Jacobian Method (Part 2)

The Jacobian Determinant & Joint Density

The joint density of (U, V) is given by:

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) \cdot |J|,$$

where J is the **Jacobian determinant** of the inverse transformation:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

Finding the Marginal Density

To find the marginal density of U, we integrate out the auxiliary variable v:

Example 9: Jacobian Method for Sum (Question)

Example

Let (X, Y) have joint density $f_{X,Y}(x, y)$. Use the Jacobian method to find the density of $Z = X + Y$.

Example 9: Jacobian Method for Sum (Solution Part 1)

Analysis: Introduce an auxiliary variable $W = X$.
The transformation is:

$$Z = X + Y, \quad W = X.$$

The inverse mapping is:

$$X = W, \quad Y = Z - W.$$

Step 1: Compute the Jacobian

$$J = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1.$$

Therefore, $|J| = |-1| = 1$.

Example 9: Jacobian Method for Sum (Solution Part 2)

Step 2: Joint Density of (Z, W)

Using the Jacobian formula, the joint density of (Z, W) is:

$$f_{Z,W}(z, w) = f_{X,Y}(w, z - w) \cdot |J| = f_{X,Y}(w, z - w) \cdot 1.$$

Step 3: Marginal Density of Z

Integrate out the auxiliary variable w:

$$f_Z(z) = \int_{-\infty}^{+\infty} f_{X,Y}(w, z - w)\, dw.$$

Changing the dummy variable w to x, we get:

$$f_Z(z) = \int_{-\infty}^{+\infty} f_{X,Y}(x, z - x)\, dx.$$

Example 10: Jacobian Method for Quotient (Question)

Example

Let (X, Y) have joint density $f_{X,Y}(x, y)$ with $Y > 0$. Use the Jacobian method to find the density of $Z = X/Y$.

Example 10: Jacobian Method for Quotient (Solution Part 1)

Analysis: Introduce an auxiliary variable $W = Y$.
The transformation is:

$$Z = \frac{X}{Y}, \quad W = Y.$$

The inverse mapping is:

$$X = ZW, \quad Y = W.$$

Step 1: Compute the Jacobian

$$J = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix} = \begin{vmatrix} w & z \\ 0 & 1 \end{vmatrix} = w \cdot 1 - z \cdot 0 = w.$$

Since $W = Y > 0$ (assuming Y is positive), we have $|J| = w$.

Example 10: Jacobian Method for Quotient (Solution Part 2)

Step 2: Joint Density of (Z, W)

Using the Jacobian formula, the joint density of (Z, W) is:

$$f_{Z,W}(z, w) = f_{X,Y}(zw, w) \cdot |J| = f_{X,Y}(zw, w) \cdot w.$$

Remark: This gives us the full joint distribution of the ratio Z and the auxiliary variable W. Next, we need to eliminate W to find the marginal distribution of Z.

Example 10: Jacobian Method for Quotient (Solution Part 3)

Step 3: Marginal Density of Z

Integrate out the auxiliary variable w:

$$f_Z(z) = \int_0^{+\infty} w \cdot f_{X,Y}(zw, w)\, dw.$$

Special Case: Independence

If X and Y are independent, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$, then:

$$f_Z(z) = \int_0^{+\infty} w f_X(zw) f_Y(w)\, dw.$$

Conclusion: This matches the quotient formula derived earlier.

Distribution of Maximum and Minimum

Let X and Y be independent with CDFs F_X and F_Y. Define $M = \max(X, Y)$ and $N = \min(X, Y)$.

CDF of Maximum

$$F_M(z) = P(\max(X, Y) \leq z) = P(X \leq z, Y \leq z) = F_X(z)F_Y(z).$$

CDF of Minimum

$$F_N(z) = 1 - P(\min(X, Y) > z) = 1 - P(X > z, Y > z) = 1 - [1 - F_X(z)][1 - F_Y(z)].$$

If continuous, the densities are:

$$f_M(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z), \quad f_N(z) = f_X(z)[1 - F_Y(z)] + f_Y(z)[1 - F_X(z)].$$

Example 11: Max and Min of Exponentials (Question)

Example

Let X, Y be independent $\text{Exp}(\lambda)$. Find the densities of $M = \max(X, Y)$ and $N = \min(X, Y)$.

Example 11: Max and Min of Exponentials (Solution)

Analysis: The CDF is $F(z) = 1 - e^{-\lambda z}$ for $z > 0$.

Solution for Maximum M

$$F_M(z) = (1 - e^{-\lambda z})^2 \implies f_M(z) = 2\lambda e^{-\lambda z}(1 - e^{-\lambda z}), \quad z > 0.$$

Solution for Minimum N

$$F_N(z) = 1 - (e^{-\lambda z})^2 = 1 - e^{-2\lambda z} \implies f_N(z) = 2\lambda e^{-2\lambda z}, \quad z > 0.$$

Thus, $N \sim \text{Exp}(2\lambda)$.

Extension to n Variables and Reliability

General n

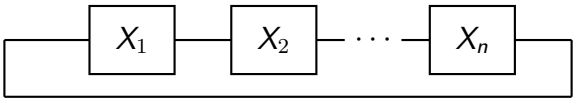
If $X_1, \ldots, X_n$ are independent with CDF $F(x)$, then:

$$F_{\max}(z) = [F(z)]^n, \quad F_{\min}(z) = 1 - [1 - F(z)]^n.$$

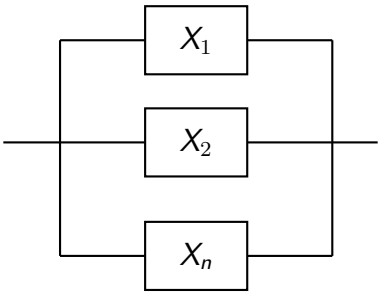
Physical Meaning in Reliability:

- **Minimum = Series System:** If a system works only if **all** components work, its lifetime is the minimum. If $X_i \sim \text{Exp}(\lambda)$, the system lifetime is $\text{Exp}(n\lambda)$.
- **Maximum = Parallel System:** If a system works if **at least one** component works, its lifetime is the maximum.

Circuit Diagram: Series vs. Parallel



Series System: $N = \min(X_1, \ldots, X_n)$



Parallel System: $M = \max(X_1, \ldots, X_n)$

Summary of Section 4

- **Discrete:** Sum probabilities of $g(x_i, y_j) = z_k$. Use Vandermonde identity for sums of Binomials.
- **Continuous CDF Method:** $F_Z(z) = \iint_{D_z} f(x,y)dxdy$, then differentiate.
- **Convolution:** $f_{X+Y}(z) = \int f_X(x)f_Y(z-x)dx$.
- **Jacobian Method:** $f_{U,V}(u,v) = f_{X,Y}(h_1, h_2)|J|$.
- **Max/Min:** $F_{\max} = F_X F_Y$, $F_{\min} = 1 - (1 - F_X)(1 - F_Y)$. Series/Parallel systems.